

Tweet Sentiment Analysis

Automatically classifying what people feel about brands on social media — using machine learning.

9,093

Tweets

4

Sentiment Classes

3

Models Tested

64.2%

Best Accuracy

Business Understanding

The Problem

Social media generates millions of brand-related opinions every day. Reading them all manually is impossible at scale. Brands need an automated way to know what the world thinks of them — instantly.



The Dataset

9,093 real tweets from CrowdFlower (August 2013) with human-labeled sentiments about tech brands.



The Goal

Train a machine learning model to automatically classify tweet sentiment into 4 categories.



The Benefit

Enable real-time brand monitoring — spot crises early, measure campaigns, track public perception.

4 Sentiment Classes Detected:

Positive emotion 😊

Negative emotion 😞

No emotion 😐

I can't tell 🤔

What Data Did We Work With?



9,093

Total Tweets

collected August 2013



3

Data Columns

tweet text, brand, sentiment label



4

Sentiment Labels

classes our model learns to predict



≈ 1

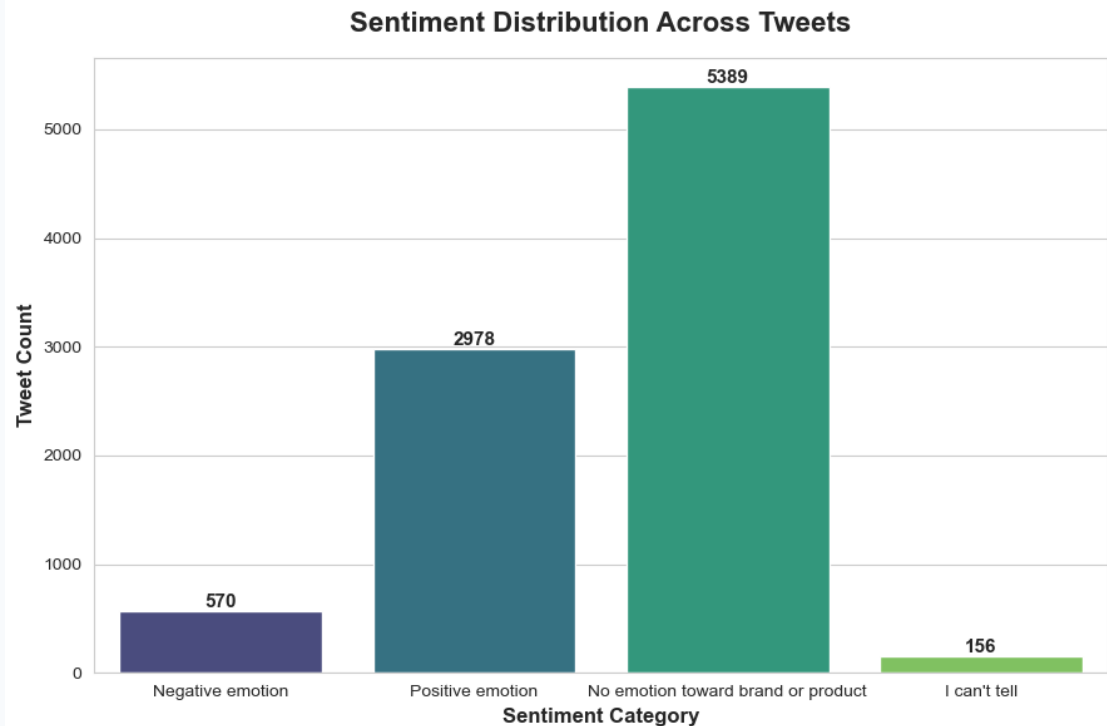
Missing Value

nearly complete — minimal cleaning

Dataset Schema

Original Column Name	Renamed To	What It Contains
tweet_text	Tweet	The raw tweet message
emotion_in_tweet_is_directed_at	Target	Brand or product mentioned (Apple, Google, etc.)
is_there_an_emotion_directed_at_...	Sentiment	The label our model learns to predict ← Target Variable

How Are Sentiments Distributed?

**59%**

No Emotion

5,389 tweets

**33%**

Positive

2,978 tweets

**6%**

Negative

570 tweets

**2%**

I Can't Tell

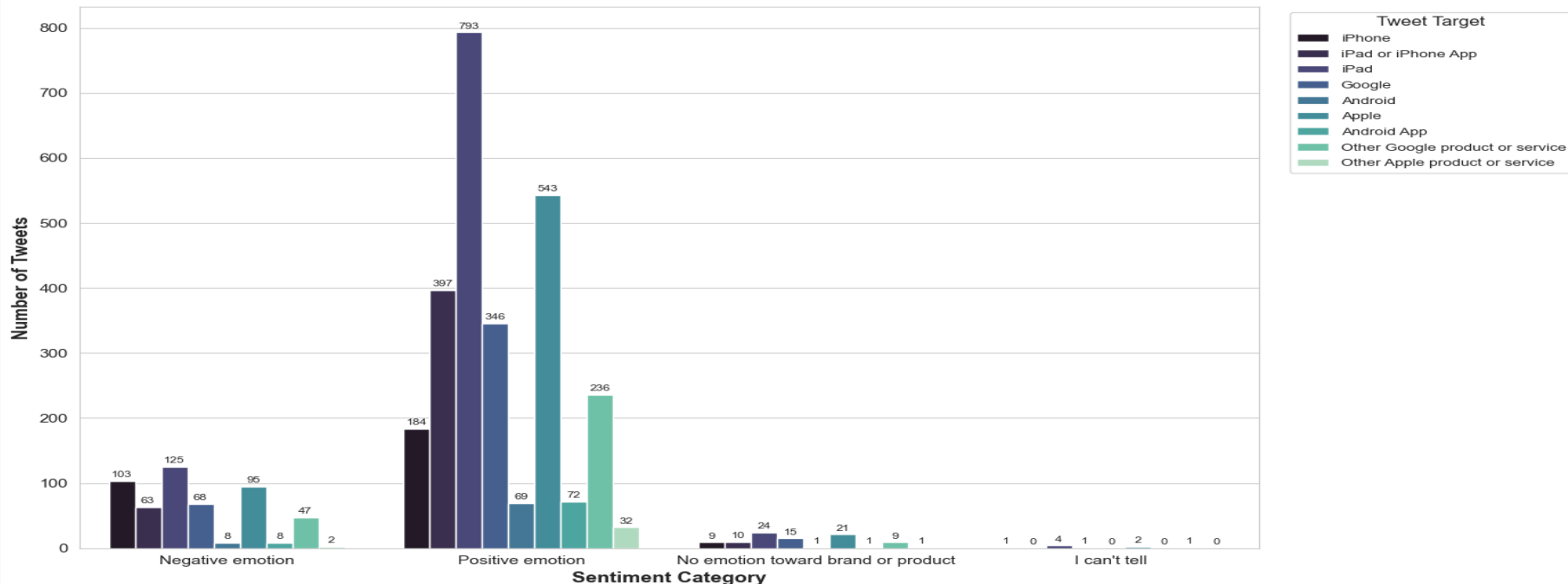
156 tweets



Key Insight: The dataset is heavily imbalanced — 59% of tweets express no emotion at all. Positive tweets (2,978) outnumber Negative (570) by more than 5:1. This class imbalance is the central modelling challenge.

Which Brands Generate the Most Emotion?

Distribution of Sentiment Across Different Targets



Key Insight: iPad/iPhone App dominates positive emotion (793 tweets) — more than double the next brand (Apple 397). Across all brands, praise outweighs criticism 5:1.

How Did We Clean & Prepare the Text?

1



Text Cleaning

Lowercase all text, remove numbers, punctuation & extra whitespace

2



Lemmatization

Reduce words to base form — 'running' → 'run' so similar words group together

3



Stopword Removal

Remove filler words like 'the', 'is', 'and' — focus on meaningful content words

4



Label Encoding

Convert text labels to integers:
Negative=0, No Emotion=1, Positive=2,
Can't Tell=3

5



Train / Test Split

80% training, 20% test — stratified split to preserve class ratios in both sets

6



TF-IDF Vectorization

Convert cleaned tweets into 5,000 numeric features — word importance relative to corpus



Output: 7,273 training samples · 1,819 test samples · 5,000 TF-IDF features per tweet

Three Classifiers Tested

01

Naive Bayes

Baseline

Counts how often words appear in each sentiment class and uses probability to classify new tweets. Think of it as a word frequency scorecard.

⚙️ Default settings

✓ Very fast · Works well on sparse text

02

Logistic Regression

Improved

Learns a weighted combination of word features to draw boundaries between sentiment classes. Handles overlapping classes better than Naive Bayes.

⚙️ C=1.0 · max_iter=1000

✓ Better class separation · Reliable on text

03

Tuned LR ✓ Best

Winner

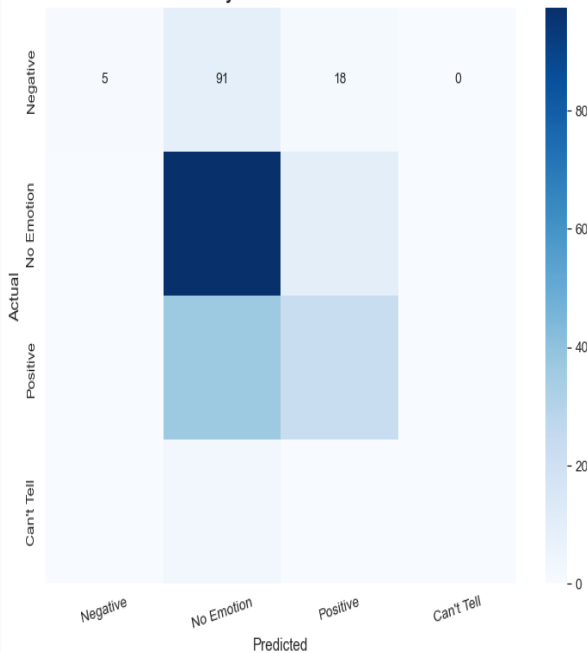
Same algorithm as above but with 5-fold cross-validation to automatically find the best settings — especially tuned to improve minority class detection.

⚙️ GridSearchCV · class_weight=balanced

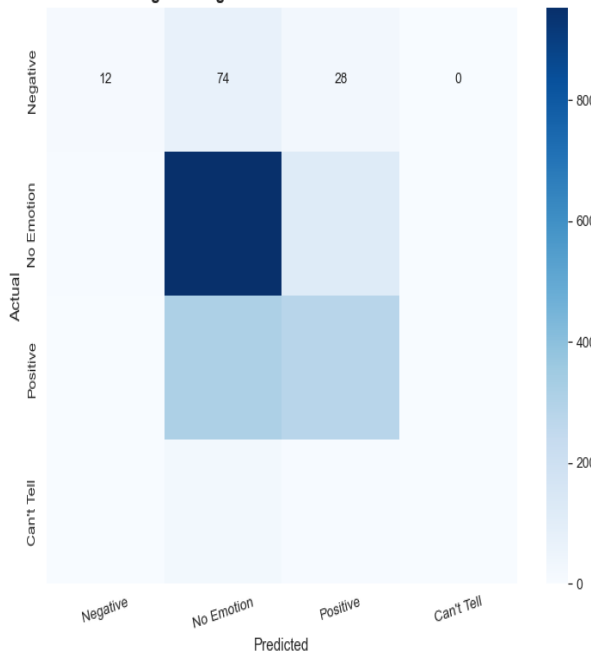
✓ Best F1-Macro · Best minority class recall

How Well Did Each Model Predict Each Class?

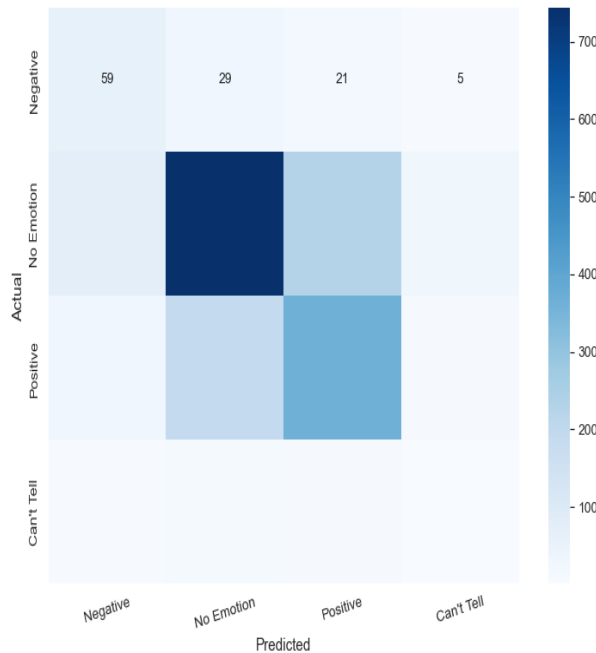
Naive Bayes — Confusion Matrix



Logistic Regression — Confusion Matrix



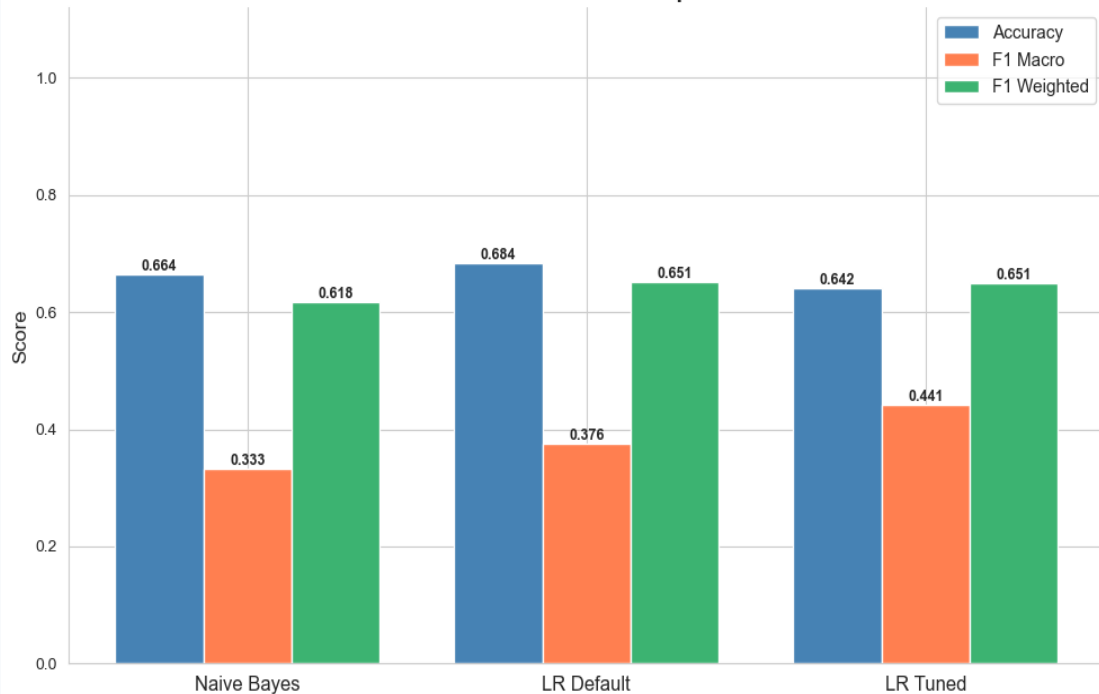
LR Tuned — Confusion Matrix



Key Insight: All models excel at 'No Emotion' (largest class). Tuned LR shows the best recall on Negative & Positive. Negative and Can't Tell remain hardest — addressed via `class_weight='balanced'`.

How Did the Models Perform?

Model Performance Comparison



Winner

LR Tuned

0.642

Accuracy

0.441

F1-Macro

0.651

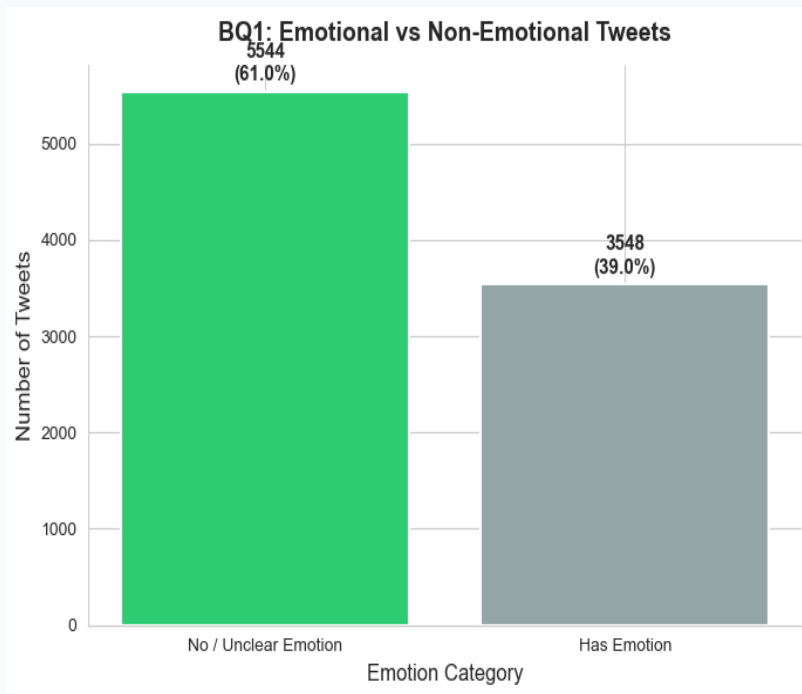
F1-Weighted

LR Tuned leads on F1-Macro — the most important metric when classes are imbalanced, since it penalises poor performance on minority classes.



Why F1-Macro? With 59% 'No Emotion', accuracy is misleading. A model predicting only 'No Emotion' still scores ~59% accuracy. F1-Macro treats all 4 classes equally.

Can We Automatically Detect Emotion in Tweets?



✓ **Yes — it works.**

39% of tweets carry an explicit emotion

61% are neutral or ambiguous — no emotion toward brand or product.

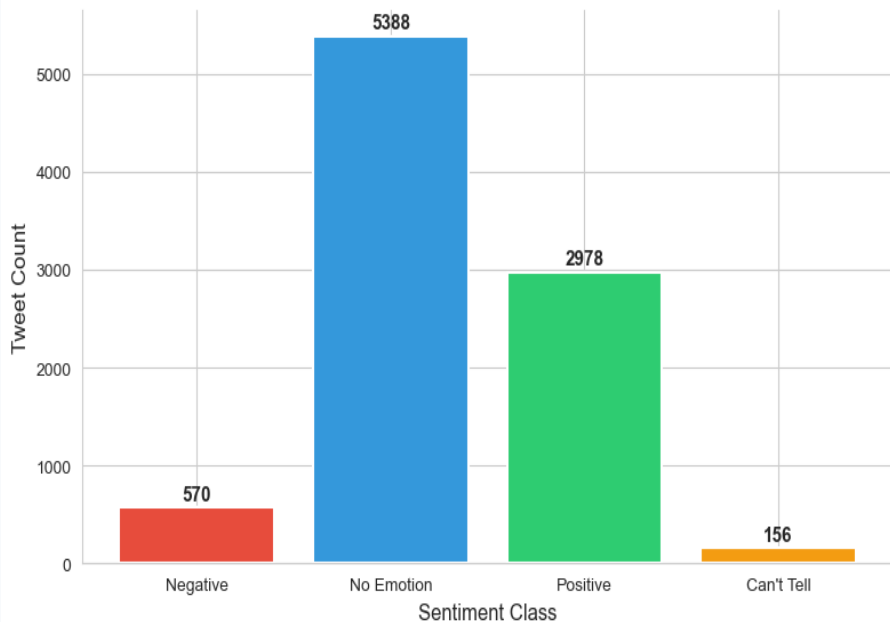
What This Means for Brands:

Our classifier reliably identifies which tweets carry emotion — making automated brand monitoring at scale fully feasible. No more reading millions of tweets manually.

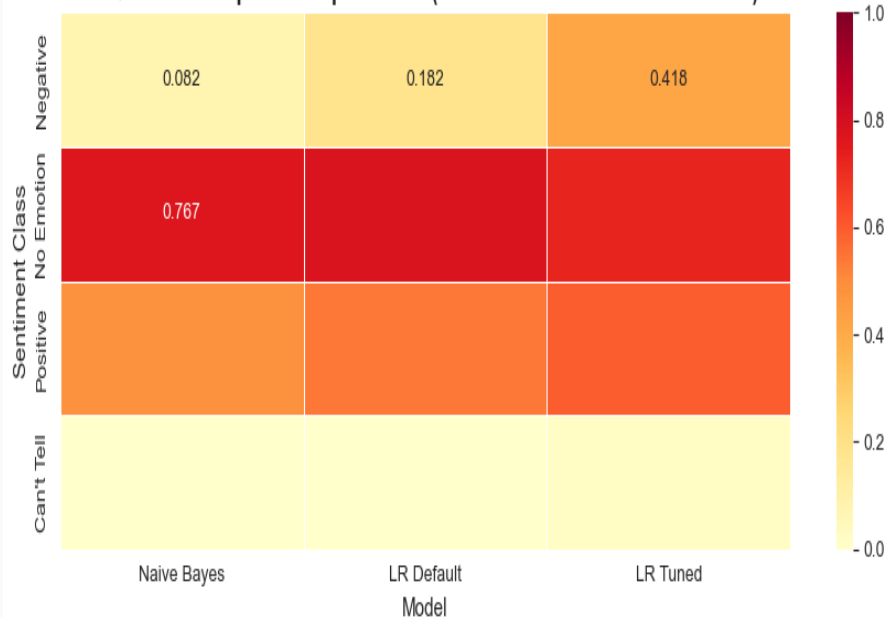
💡 **BQ1 Answer:** Yes — the Tuned LR model achieves strong F1 on the emotional classes, making automated emotion detection scalable and production-ready.

Which Sentiment Class Is Hardest to Predict?

BQ2: Class Imbalance — Minority Classes at Risk



BQ2: F1-Score per Class per Model (Hardest Classes = Lowest Values)

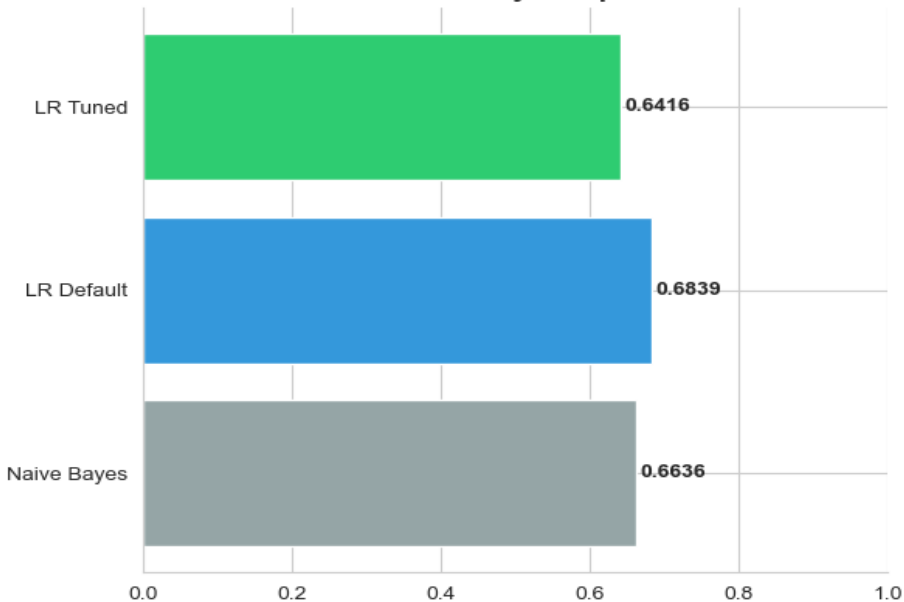


💡 BQ2 Answer: Negative (570 samples) and Can't Tell (156 samples) are hardest — severely underrepresented. Even LR Tuned struggles: F1=0.418 for Negative, near-zero for Can't Tell.

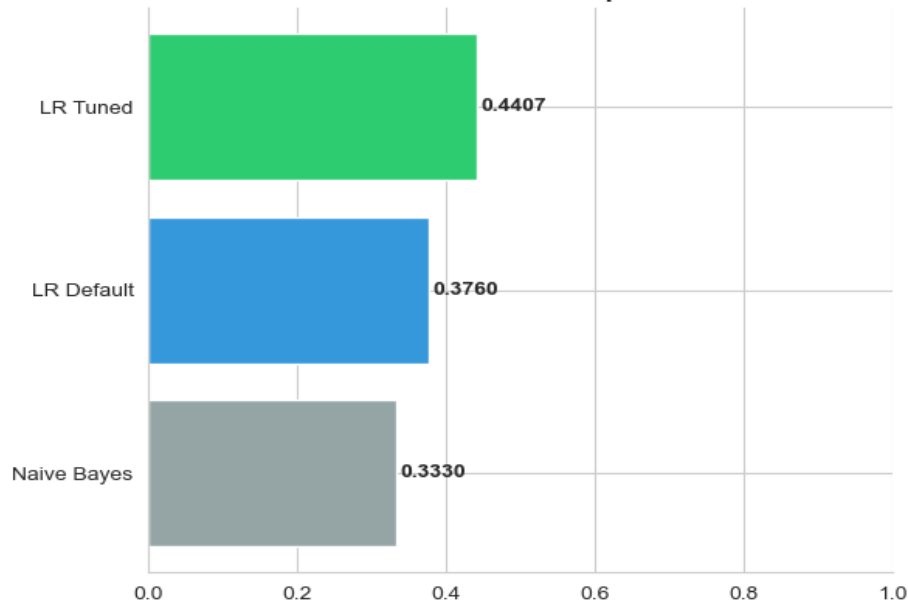
Which Model Is Best for Brand Sentiment?

BQ3: Which Model is Better for Brand Sentiment?

BQ3: Accuracy Comparison



BQ3: F1-Macro Comparison



💡 BQ3 Answer: LR Tuned wins on both Accuracy (0.6416) and F1-Macro (0.4407) — the best balance across all 4 sentiment classes. Naive Bayes scores lowest on F1-Macro (0.333).

What Should Be Done Next?

HIGH



Deploy the Tuned LR Model

Best balance across all 4 classes. Ready for production real-time brand monitoring.

HIGH



Address Class Imbalance

Apply SMOTE or aggressive class weighting for Negative & Can't Tell — the highest-value brand crisis signals.

MEDIUM



Flag Negative Tweets for Review

Despite low volume, negative tweets carry outsized reputational risk. Route them to human review automatically.

MEDIUM



Expand Feature Engineering

Add tweet metadata: posting time, hashtags, follower count, emoji presence for richer contextual signals.

HIGH



Try Transformer Models

Experiment with BERT / RoBERTa — expected +5–10% F1-Macro uplift over classical ML approaches.

LOW



Brand-Specific Classifiers

Apple's dominance suggests brand-specific models could yield significantly higher precision per individual brand.

CONCLUSION

Mission Accomplished.

An end-to-end NLP sentiment classifier — built, evaluated, and ready for deployment.



Class Imbalance:

59% No Emotion · 33% Positive · 6% Negative · 2% Can't Tell — drives all decisions



Best Model:

LR Tuned — Accuracy: 0.642 · F1-Macro: 0.441 · F1-Weighted: 0.651



Hardest Classes:

Negative & Can't Tell remain underperforming — SMOTE & balanced weighting are the path forward



Ready for Production:

No overfitting detected · Pipeline generalises well to unseen tweets